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PHANTOM POSITIONS

IDENTIFYING GHOST JOB POSTINGS IN ONLINE LISTINGS

- FINAL REPORT

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REMOTE REPOSITORY
HTTPS://GITHUB.COM/IANRPAGE/PHANTOMPOSITIONS

CAP4770 – Introduction To Data Science
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Introduction/Executive Summary

The proliferation of online job platforms has created significant opportunities for both employers and job seekers; however, it has also introduced a vector for fraud. Fraudulent job postings (listings designed to extract personal information, solicit fees, or otherwise deceive applicants) represent a growing threat to vulnerable populations, including recent graduates, career changers, and the unemployed. The consequences of engaging with such postings range from financial loss to identity theft.

To address the problem of fraudulent job listings and to support automated detection of such, this project, ***Phantom Positions***, applies supervised classification and text mining techniques to a publicly available, labeled dataset. Three classification models, Support Vector Machine (SVM), Random Forest (RF), and Logistic Regression (LR), are evaluated across three feature conditions: text only, metadata only, and combined text and metadata. The strongest overall fraud-class F1 score was achieved by SVM on the combined feature set (0.838), while RF demonstrated a decisive advantage in the metadata-only condition, highlighting meaningful differences in how each model uses available feature types.

Improvements After the Last Presentation

Following the progress update presentation, two key changes were made. First, the preprocessing pipeline was refactored to move TF-IDF vectorization from $build_features()$ into $load_and_split()$, and the vectorizer was updated to include additional parameters for improved text representation. Second, LR was added as a third classification model, enabling a more complete comparison of linear and ensemble approaches across all feature conditions.

Literature Review

Prior work on this problem has largely converged on the Employment Scam Aegean Dataset (EMSCAD). Vidros et al. (2017) [1] introduced the dataset and established the first classification baselines using bag of words features across six classifiers, with Random Forest achieving an F-measure of 0.912 on a balanced subsample. Critically, when tested on the full dataset, fraudulent class precision dropped to 0.282, directly motivating our use of F1-score and resampling strategies. Amaar et al. (2022) [2] subsequently demonstrated that combining TF-IDF text features with structured metadata outperforms either feature type alone. Bhatia and Meena (2023) [3] applied TF-IDF with SMOTE and ADASYN oversampling to address class imbalance, providing methodological precedent for our approach. Pillai (2023) [4] represents the current state of the art, employing a Bi-LSTM deep learning model, establishing a performance ceiling against which our interpretable classification models can be contextualized.

This project uses the "Real or Fake Job Posting Prediction" dataset, sourced from the EMSCAD [1] and available publicly on Kaggle[1]. The dataset contains 17,880 job postings labeled for binary classification (legitimate or fraudulent) across 18 features. Structured metadata features include binary indicators (company logo presence, screening questions, telecommuting status) and nominal fields (employment type, required experience, required education, industry, and function). Free-text features include the job title, location, company profile, job description, requirements, and benefits.

The dataset is highly imbalanced at approximately 95% legitimate and 5% fraudulent, which makes standard accuracy an inadequate evaluation metric. A classifier predicting every posting as legitimate would achieve 95% accuracy while detecting no fraud whatsoever. F1-score is therefore selected as the primary metric, as it accounts for both precision and recall and is sensitive to performance on the minority class.

[1] https://www.kaggle.com/datasets/shivamb/real-or-fake-fake-jobposting-prediction/data

Modeling

This project is implemented in Python using pandas, scikit-learn, imbalanced-learn, matplotlib, and seaborn within a Jupyter Notebook environment, with version control managed via GitHub. Our simple end-to-end system diagram is provided in Appendix A, and its content is described in detail in the following sections.

Preprocessing and Feature Engineering

A shared preprocessing pipeline ($preprocessing.py$) was developed to ensure consistency across all models and conditions. Free-text fields were concatenated and represented as TF-IDF weighted vectors using $stop_words = 'english'$, $ngram_range = (1,2)$, $min_df = 5$, $max_df = 0.9$, and $max_features = 5000$ for cross-model comparisons. Structured metadata features were encoded using binary and one-hot encoding. The combined condition concatenates TF-IDF and metadata representations into a unified sparse feature matrix using $scipy.sparse.hstack$. Features were scaled using $StandardScaler$ with $with_mean = False$ to preserve sparsity. The dataset was split into training and test sets using a 70/30 stratified split ($test_size = 0.3$, $stratify = Y$), preserving the original class distribution in both splits. This produces a test set of approximately 5,104 legitimate and 260 genuine fraudulent postings (5,364 total), reflecting the real-world imbalance. SMOTE ($k_neighbors = 3$) was applied exclusively to the training set to address class imbalance without introducing leakage into evaluation. $random_state = 5$ was set consistently across all splitting, sampling, and model steps to ensure reproducibility.

Classification Models

Three classification models were evaluated across three feature conditions: text features only, metadata features only, and combined text and metadata features.

Support Vector Machine: Implemented using scikit-learn's $LinearSVC$, SVMs classify data by finding the optimal hyperplane that maximizes the margin between the closest data points (the support vectors) of different classes. They are well-suited to high-dimensional sparse feature spaces such as those produced by TF-IDF. The key parameter is $C = 0.01$, which defines the strength of regularization (inversely proportional to C), applying strong regularization to mitigate overfitting in the sparse, high-dimensional feature space and balance bias-variance tradeoff.

Random Forest: Implemented using scikit-learn's $RandomForestClassifier$, RF builds an ensemble of decision trees, utilizing a random sample of data which helps to reduce bias and overfitting. It works by first creating many decision trees using random data, then picking random features to decide how to split the data. Each tree makes a prediction based on what it learned, and the final prediction is made based on the majority vote. Parameters were set to $n_estimators = 200$; the increase from 100 to 200 trees was identified through manual tuning as the single most effective improvement. Random search was also explored across class weight, bootstrap, max depth, and minimum samples parameters, but results were mixed: sometimes reducing precision while improving recall, other times showing little change from baseline. Ultimately, the default configuration with increased tree count provided the best results.

Logistic Regression: Implemented using scikit-learn's $LogisticRegression$, LR models the probability of a binary outcome using a logistic function to map linear predictions to probabilities between 0 and 1, typically applying a 0.5 threshold to classify results. It works well with wide, sparse feature matrices such as those produced by TF-IDF. The parameter $C = 0.01$ applies high regularization, penalizing the model for assigning large weights to any single feature, which allows the model to generalize better and avoids overfitting. The $class_weight = \{0: 1, 1: 6\}$ parameter makes the model treat every fraudulent posting as five times more important than a legitimate one during training, counteracting the model's natural tendency to label a posting as legitimate given the class imbalance. Although cross-model comparisons use $max_features = 5000$, individual tuning of the LR model used $max_features = 3000$ within the TF-IDF vectorization and was found to strengthen recall on the fraud class.

Metrics & Evaluation Setup

All models were evaluated on the same held-out test set described in the Preprocessing section above. Evaluation used scikit-learn's $classification_report$, roc_auc_score, and $average_precision_score$.

F1-score on the fraud (minority) class is the primary metric, as it balances precision and recall for the class of practical interest. Raw accuracy is insufficient given the ~95/5 class split, a classifier predicting every posting as legitimate would achieve 95% accuracy while detecting no fraud whatsoever. ROC AUC and Average Precision (AP) are reported as threshold-independent measures of discriminative ability, both of which are robust to class imbalance. Confusion matrices are provided in the Appendix for each model and condition to make false negative and false positive rates explicit, as both carry real costs in a fraud detection context: false negatives allow fraudulent postings to go undetected, while false positives incorrectly flag legitimate listings.

Results

The following abbreviations are used throughout: TN (true negatives), FP (false positives), TP (true positives), FN (false negatives), ROC AUC (Receiver Operating Characteristic Area Under the Curve), and AP (Average Precision). Full per-condition confusion matrices and ROC curves for all models are provided in the Appendix.

Support Vector Machine

The SVM's strongest results were on the combined feature condition, where it achieved the highest fraud-class F1 of any model. The model achieved TN=5077, FP=27, TP=207, FN=53, with fraud class precision=0.88, recall=0.80, F1=0.84, ROC AUC=0.968, and AP=0.869, with overall accuracy of 0.99. The model did well considering the severe class imbalance, high-dimensional sparse features from TF-IDF, and the noisy text data in job postings, successfully detecting 80% of phantom postings with high precision. SVM's efficiency in high-dimensional sparse spaces carries through into the combined condition, where its strength on text features gives it an edge over both RF and LR despite RF's higher AUC.

Random Forest

While RF performed competitively on the combined and text-only conditions, its most distinctive result was in the metadata-only condition, where it substantially outperformed the other two models. On metadata-only, the model achieved TN=4936, FP=168, TP=182, FN=78, with fraud class precision=0.52, recall=0.70, F1=0.597, ROC AUC=0.932, and AP=0.675, compared to F1=0.386 for SVM and F1=0.275 for LR on the same condition. RF's ability to capture non-linear interactions among sparse categorical features gives it a genuine structural advantage when text is unavailable. On the combined condition, RF achieved TN=5100, FP=4, TP=176, FN=84, F1=0.80, and the highest ROC AUC of any model (0.988), though this did not translate into the best operating-threshold performance. Overall, the results suggest that the preprocessing pipeline and default values can lead to the best performance; further tuning offered diminishing returns.

Logistic Regression

Logistic regression achieves higher recall with a lower number of maximum features from the TF-IDF vectorizer; although results below use $max_features = 5000$ for cross-model comparability, individual tuning used $max_features = 3000$. (See Appendix D.ii. for results.) The class weight parameter makes the model treat every fraudulent posting as six times more important than a legitimate one during training, deliberately biasing the model toward flagging suspicious listings rather than missing them. High regularization penalizes the model for assigning large weights to a single feature, preventing over-reliance on any one of the features from the vectorizer.

On the combined condition (the most meaningful for LR) the model achieved the highest recall of any model (0.819), with TN=5048, FP=56, TP=213, FN=47, precision=0.79, F1=0.805, ROC AUC=0.979, and AP=0.881. LR is better than the other models at detecting true positives at the cost of a higher number of false positives, a result of the model's hyperparameter tuning causing it to be more sensitive in its predictions. While this comes at the cost of more false positives, incorrectly flagging legitimate postings is less harmful than allowing a ghost posting to go undetected. The metadata-only condition reveals the limits of this strategy: high recall (0.89) is achieved at the cost of 1,234 false positives, making the model impractical without text features.

Conclusion

Cross-model comparison across the three feature conditions reveals distinct patterns in how each classifier uses available information.

Combined features produced the strongest results across all models, confirming the prior finding by Amaar et al. [2] that text and metadata features are complementary. Our original hypothesis anticipated RF as the top performer on the combined condition; this did not hold. SVM achieved the highest fraud-class F1 (0.838), with LR (0.805) and RF (0.800) close behind. As shown in Figure 1 below, the spread across all three models is narrow, and RF's higher ROC AUC (0.988 vs. 0.968 for SVM) did not translate into better performance at the operating threshold.

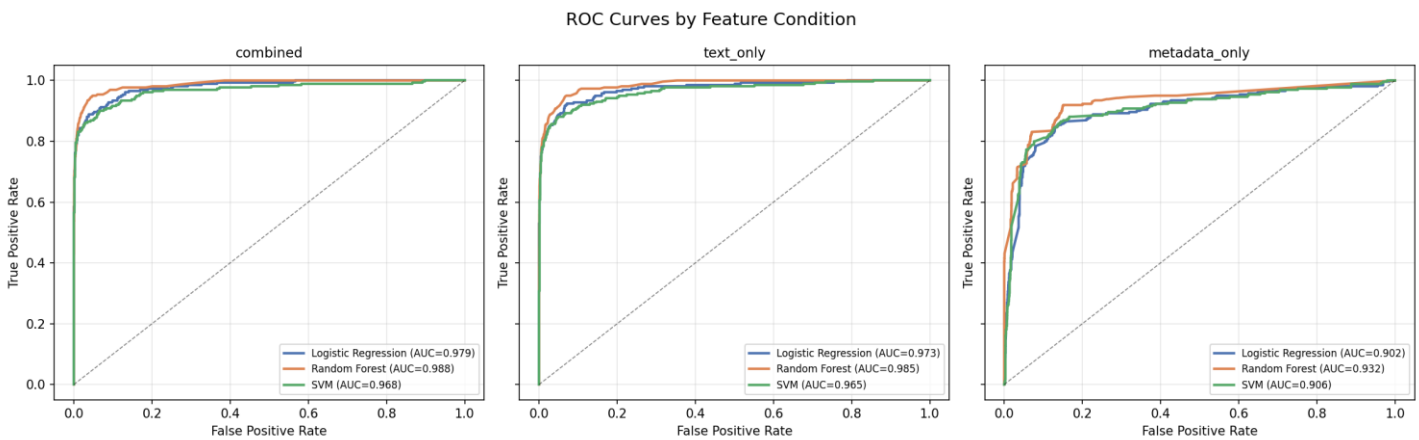

Figure 1: ROC Curves by Feature Condition

Text-only results were similarly tight: SVM 0.793, RF 0.780, LR 0.770. The negligible gap across models suggests that TF-IDF representations are sufficiently informative that classifier choice has little bearing on outcome: all three extract roughly equivalent signal from text features alone. This similarity is shown in Figure 2, below.

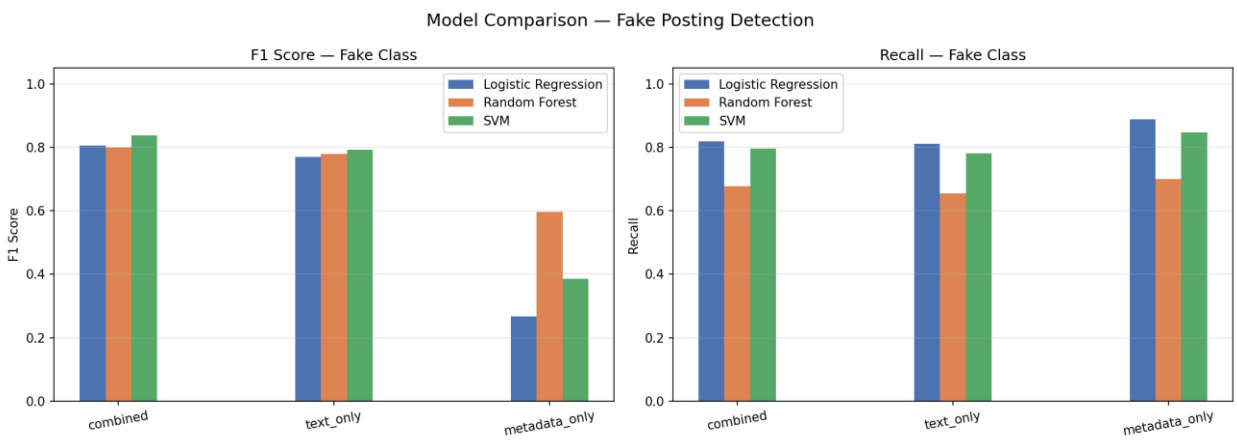

Figure 2: F1 and Recall Bar Charts — Model Comparison

Metadata-only is the most theoretically informative condition. RF's F1 of 0.597 is substantially higher than SVM (0.386) and LR (0.275), and the separation is meaningful rather than marginal. Logistic regression's precision collapses to 0.16 in this condition, reflecting the limits of a linear model and aggressive class weighting when features are sparse categorical indicators rather than rich text. RF's ability to capture non-linear interactions among structured features gives it a genuine structural advantage when text is unavailable, which is a relevant consideration for real-world deployment scenarios where job postings may be incomplete.

Across all conditions, the metadata-only results for all models underscore that text is the primary driver of classification performance in this dataset. The combined condition's modest improvement over text-only suggests metadata contributes meaningful signal at the margin but is not independently sufficient. Figure 3 highlights this effect across models.

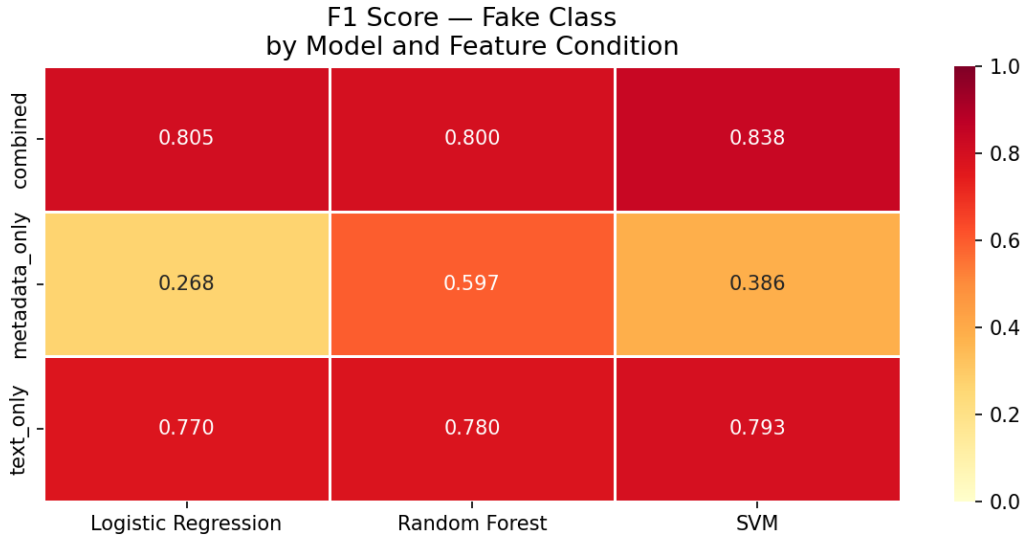

Figure 3: Heatmap — Fake Class by Model and Feature Condition

Taken together, these findings suggest that a production screening tool would be best served by SVM or LR on combined features for balanced fraud detection, with RF as the preferred fallback when text content is absent or unreliable. Several avenues for future work emerge from this project. Threshold calibration could reduce false positives across all three models without sacrificing recall: a particularly meaningful improvement for LR and SVM in deployment contexts where flagging legitimate postings carries real costs. K-fold cross-validation would provide more robust performance estimates by training each model multiple times and averaging results, reducing sensitivity to any particular train/test split. All models, particularly SVM, would benefit from substantially more training data; SVM's low variance means it is well-positioned to estimate a more stable decision boundary given a larger sample, and the same holds for improving minority-class representation across all classifiers.

Most significantly, the dataset itself represents a scalability constraint that future work should address directly. Collected prior to the widespread adoption of remote work, the EMSCAD dataset reflects a labor market in which work-from-home listings were relatively uncommon. In the post-COVID era, remote and hybrid job postings have grown, and with them, the likely scale and sophistication of fraudulent listings. A dataset that captures this shift would not only provide more training samples but would better represent the current fraud landscape, potentially revealing new linguistic and structural patterns not present in the existing data. Extending this evaluation to a more recent dataset, or augmenting the EMSCAD with post-2020 postings, would be a meaningful contribution to the field.

References

[1] Vidros, S., Kolias, C., Kambourakis, G., & Akoglu, L. (2017). Automatic detection of online recruitment frauds: Characteristics, methods, and a public dataset. *Future Internet, 9*(1), 6. https://doi.org/10.3390/fi9010006

[2] Amaar, A., Aljedaani, W., Rustam, F., Ullah, S., Rupapara, V., & Ludi, S. (2022). Detection of fake job postings by utilizing machine learning and natural language processing approaches. *Neural Processing Letters, 54*(3), 2219-2247. https://doi.org/10.1007/s11063-021-10727-z

[3] Bhatia, R., & Meena, G. (2023). Detection of fake online recruitment using machine learning techniques. *IEEE Conference Publication.* https://ieeexplore.ieee.org/document/10074276

[4] Pillai, A. S. (2023). Detecting fake job postings using bidirectional LSTM. *International Research Journal of Modernization in Engineering Technology and Science, 5*(3). https://doi.org/10.56726/IRJMETS35202

Appendix

A. System/Workflow Diagram

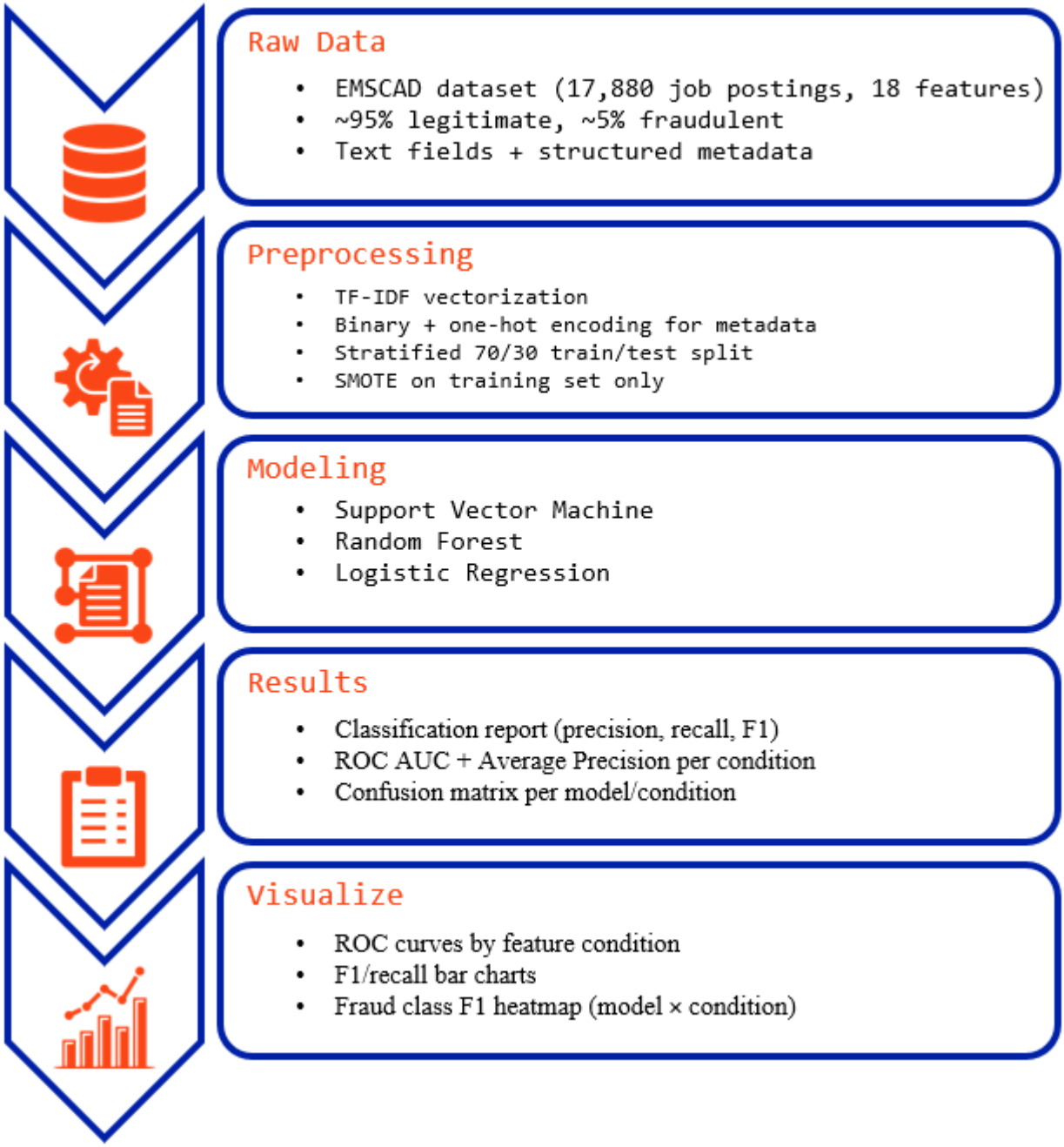

Figure 1: End-to-End System Diagram

B. SVM

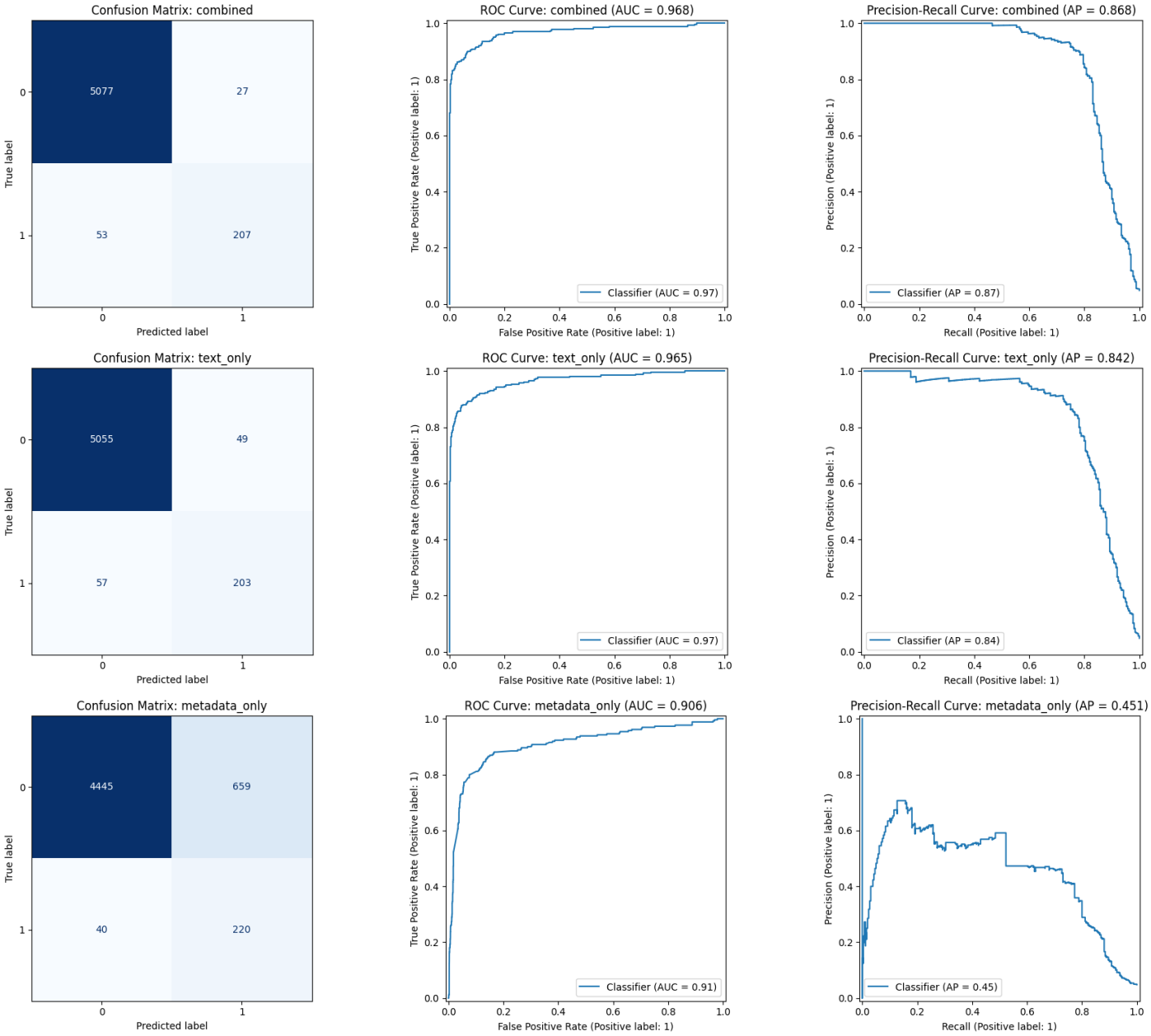

Figure 2: SVM per-condition confusion matrices / ROC curves

C. Random Forest

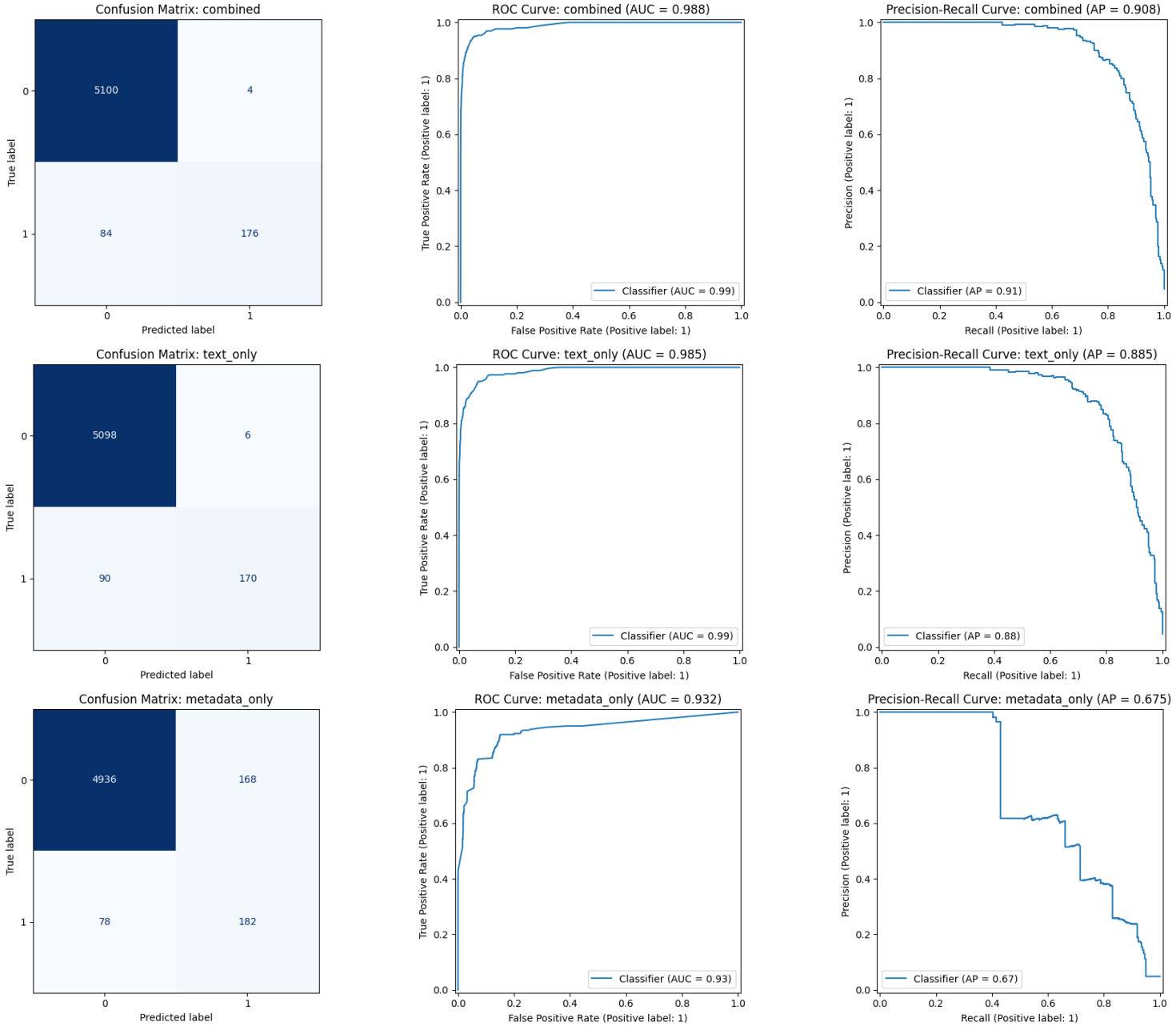

Figure 3: RF per-condition confusion matrices / ROC curves

D. Logistic Regression

i. Results using $max_features = 5000$

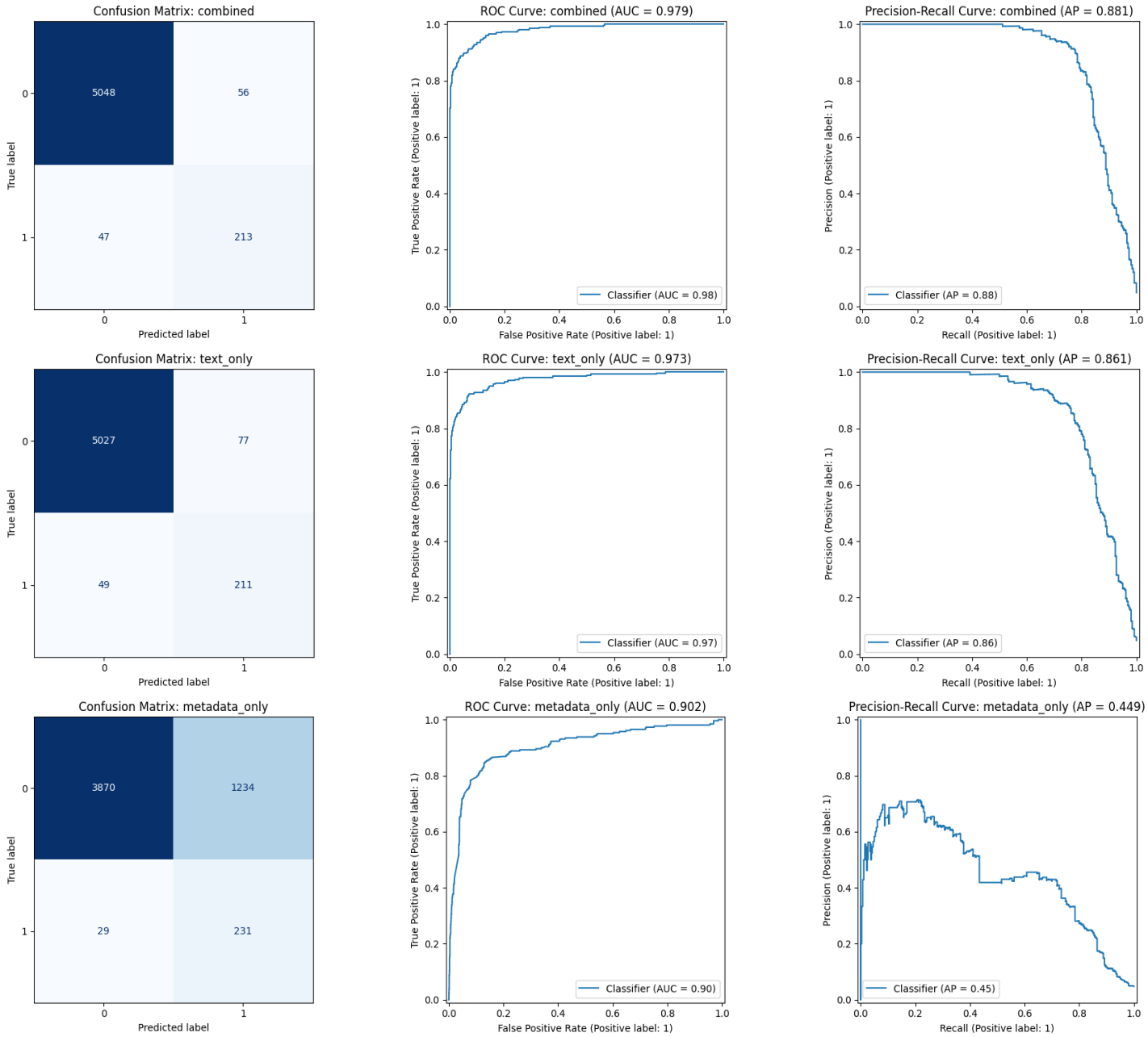

Figure 4: LR per-condition confusion matrices / ROC curves (max_features=5000)

ii. Results using $max_features = 3000$

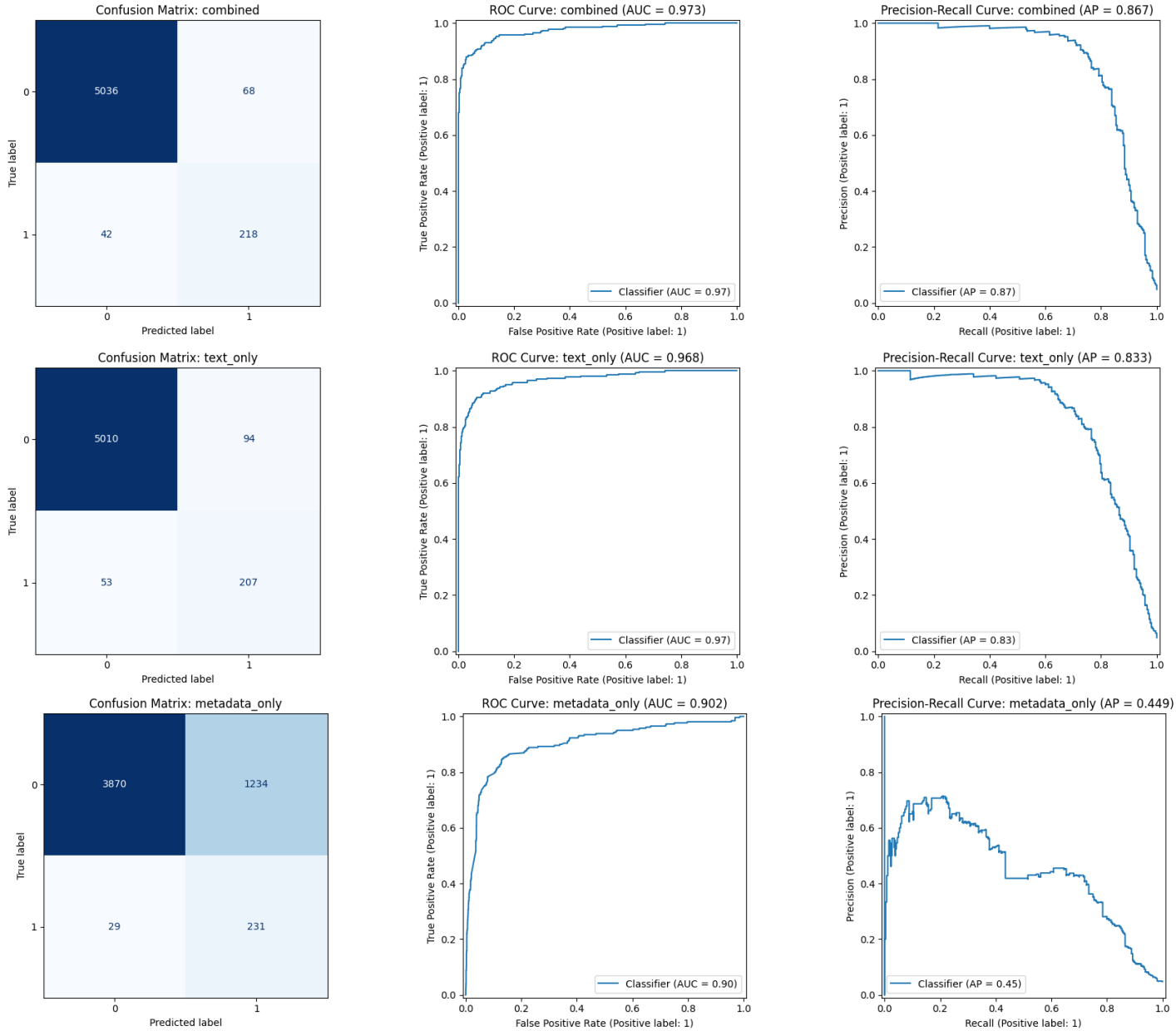

Figure 5: LR per-condition confusion matrices / ROC curves (max_features=3000)